Publications - Felipe Alarcón

Complete publication list on NASA ADS

53	9 + 5	27	2,590
Refereed publications	First / second author	h-index	Citations (NASA ADS)

Plus 8 submitted articles under review.

First-author papers

1. "EK Cha b: multi-tracer detection of a planetary-mass companion candidate in a gas-rich environment." **Alarcón**, Chowbay, Facchini et al, accepted for publication in A&A Letters. arXiv:2610.10701.
2. "exoALMA. XXIV. Formaldehyde Emission in Protoplanetary Disks of exoALMA Compared with Their Properties and Dynamical State." **Alarcón**, Facchini, Trapman et al, ApJL 1000 L32 (2026).
3. "Silicon Sulfide Isotopologues Reveal Silicate Sublimation in the Outer Disk of PDS 66" Yoshida, **Alarcón** et al, ApJL, 999, L22 (2026). *(shared first authorship)*
4. "Thermal Structure and Millimeter Emission of Protoplanetary Disk with embedded protoplanets from radiative transfer modeling" **Alarcón** & Bergin, ApJ, 967, 144 (2024).
5. "Extinction Values toward Embedded Planets in Protoplanetary Disks Estimated from Hydrodynamic Simulations" **Alarcón**, Bergin, and Cugno, ApJ, 966, 225 (2024)
6. "Viscous heating as the dominant heat source inside the water snowline of V883 Ori" **Alarcón**, Casassus, Lyra et al, MNRAS, 527, 9655 (2024).
7. "A Localized Kinematic Structure Detected in Atomic Carbon Emission Spatially Coincident with a Proposed Protoplanet in the HD 163296 Disk" **Alarcón**, Bergin, and Teague ApJL, 941, L24 (2022)
8. "Molecules with ALMA at Planet-forming Scales (MAPS). VIII. CO Gap in AS 209-Gas Depletion or Chemical Processing?" **Alarcón**, Bosman, Bergin et al. ApJS, 257, 8 (2021)
9. "Chemical Evolution in a Protoplanetary Disk within Planet Carved Gaps and Dust Rings" **Alarcón**, Teague, Zhang et al. ApJ, 905, 68 (2020)

Refereed papers · major or significant contributions

* denotes second authorship.

10. * "Dust Recycling in Protoplanetary Disks: A Novel Method of Volatile Enrichment in Embedded Giant Planets" Van Clepper, **Alarcón**, Ciesla et al. ApJL, 994, L44 (2025).
11. * "Tracking the Chemical Evolution of Hydrocarbons Through Carbon Grain Supply in Protoplanetary Disks" Raul, **Alarcón**, and Bergin ApJ, 982, 155 (2025)
12. * "CI Traces the Disk Atmosphere in the IM Lup Protoplanetary Disk" Law, **Alarcón**, Cleeves et al. ApJL, 959, L27 (2023)
13. * "Molecules with ALMA at Planet-forming Scales (MAPS). VII. Substellar O/H and C/H and Superstellar C/O in Planet-feeding Gas" Bosman, **Alarcón**, Bergin et al. ApJS, 257, 7 (2021)
14. * "Destruction of Refractory Carbon Grains Drives the Final Stage of Protoplanetary Disk Chemistry" Bosman, **Alarcón**, Zhang et al. ApJ, 910, 3 (2021)
15. "Direct Measurement of Extinction in a Planet-Hosting Gap " Cugno, Facchini, **Alarcón** et al. AJ, 170, 317 (2025)
16. "PDRs4All. VIII. Mid-infrared emission line inventory of the Orion Bar" Van De Putte, Meshaka, Trahin et al. A&A, 687, A86 (2024)
17. "A far-ultraviolet–driven photoevaporation flow observed in a protoplanetary disk" Berné, Habart, Peeters et al. Sci, 383, 988 (2024)
18. "Formation of the methyl cation by photochemistry in a protoplanetary disk" Berné, Martin-Drumel, Schroetter et al. Nature, 621, 56 (2023)

19 "PDRs4All. IV. An embarrassment of riches: Aromatic infrared bands in the Orion Bar" Chown, Sidhu, Peeters et al. A&A, 685, A75 (2024)

20 "Molecules with ALMA at Planet-forming Scales. XX. The Massive Disk around GM Aurigae" Schwarz, Calahan, Zhang et al. ApJS, 257, 20 (2021)

Refereed papers · author contributions and/or data reduction

21 "JWST-DECO: The Impact of Accretion on Mid-Infrared Observable Water in Planet-forming Disks." Calahan, Dziire, Öberg et al. ApJ, 1004,213 (2026)

22 "PDRs4All: XX. The 6 to 9μm region as a probe of PAH charge and size in the Orion Bar." Khan, Daza Rodriguez, Peeters et al. A&A, 709, A39 (2026)

23 "PDRs4All: XIX. The evolution of the PAH ionisation and PAH size distribution across the Orion Bar." Maragkoudakis, Boersma, Peeters et al. A&A, 709, A38 (2026)

24 "Tracing Pebble Drift History in Two Protoplanetary Disks with CO Enhancement" Armitage, Williams, Zhang et al. ApJ, 998, 308 (2026)

25 "JWST-MIRI Observations of the Irradiated Chemistry in the Inner Disk Cavity of GM Aur." Romero-Mirza, Öberg, Banzatti et al. ApJ, 991, 128 (2025)

26 "PDRs4All XV: CH radical and H_3^+ molecular ion in the irradiated protoplanetary disk d203-506" Schroetter, Berné, Goicoechea et al. A&A, 699, L13 (2025)

27 "PDRs4All: XIV. Probing CH out-of-plane bending modes of PAH molecules in the Orion Bar with JWST" Khan, Abbott, Peeters et al. A&A, 699, A133 (2025)

28 "A solar C/O ratio in planet-forming gas at 1 au in a highly irradiated disk" Schroetter, Berné, Bron et al. Nat Astron (2025). doi:10.1038/s41550-025-02596-6

29 "PDRs4All: XIII. Empirical prescriptions for the interpretation of JWST imaging observations of star-forming regions" Chown, Okada, Peeters et al. A&A, 698, A86 (2025)

30 "PDRs4All: XII. Far-ultraviolet-driven formation of simple hydrocarbon radicals and their relation with polycyclic aromatic hydrocarbons" Goicoechea, Pety, Cuadrado et al. A&A, 696, A100 (2025)

31 "PDRs4All: X. ALMA and JWST detection of neutral carbon in the externally irradiated disk d203-506: Undepleted gas-phase carbon" Goicoechea, Le Bourlot, Black et al. A&A, 689, L4 (2024)

32 "PDRs4All. IX. Sulfur elemental abundance in the Orion Bar" Fuente, Roueff, Le Petit et al. A&A, 687, A87 (2024)

33 "PDRs4All. VII. The 3.3 μm aromatic infrared band as a tracer of physical properties of the interstellar medium in galaxies" Schroetter, Berné, Joblin et al. A&A, 685, A78 (2024)

34 "PDRs4All. VI. Probing the photochemical evolution of PAHs in the Orion Bar using machine learning techniques" Pasquini, Peeters, Schefter et al. A&A, 685, A77 (2024)

35 "OH as a probe of the warm-water cycle in planet-forming disks" Zannese, Tabone, Habart et al. NatAs, 8, 577 (2024)

36 "Astronomical CH3+ rovibrational assignments. A combined theoretical and experimental study validating observational findings in the d203-506 UV-irradiated protoplanetary disk" Changala, Chen, Li et al. A&A, 680, A19.

37 "PDRs4All. V. Modelling the dust evolution across the illuminated edge of the Orion Bar" Elyajouri, Ysard, Abergel et al. A&A, 685, A76 (2024)

38 "PDRs4All: III. JWST's NIR spectroscopic view of the Orion Bar" Peeters, Habart, Berné et al. A&A, 685, A74 (2024)

39 "PDRs4All. II. JWST's NIR and MIR imaging view of the Orion Nebula" Habart, Peeters, Berné et al. A&A, 685, A73 (2024)

40 "Molecules with ALMA at Planet-forming Scales (MAPS): Complex Kinematics in the AS 209 Disk Induced by a Forming Planet and Disk Winds" Galloway-Sprietsma, Bae, Teague et al. ApJ, 950, 147 (2023)

41 "Molecules with ALMA at Planet-forming Scales (MAPS): A Circumplanetary Disk Candidate in Molecular-line Emission in the AS 209 Disk" Bae, Teague, Andrews et al. ApJL, 934, L20 (2022)

42 "PDRs4All: A JWST Early Release Science Program on Radiative Feedback from Massive Stars" Berné, Habart, Peeters et al. PASP, 134, 054301 (2022)

43 "Molecules with ALMA at Planet-forming Scales (MAPS). I. Program Overview and Highlights" Öberg, Guzmán, Walsh et al. ApJS, 257, 1 (2021)

44 "Molecules with ALMA at Planet-forming Scales (MAPS). XVII. Determining the 2D Thermal Structure of the HD 163296 Disk" Calahan, Bergin, Zhang et al. ApJS, 257, 17 (2021)

45 "Molecules with ALMA at Planet-forming Scales (MAPS). XV. Tracing Protoplanetary Disk Structure within 20 au" Bosman, Bergin, Loomis et al. ApJS, 257, 15 (2021)

46 "Molecules with ALMA at Planet-forming Scales (MAPS). IV. Emission Surfaces and Vertical Distribution of Molecules" Law, Teague, Loomis et al. ApJS, 257, 4 (2021)

47 "Molecules with ALMA at Planet-forming Scales (MAPS). V. CO Gas Distributions" Zhang, Booth, Law et al. ApJS, 257, 5 (2021)

48 "Molecules with ALMA at Planet-forming Scales (MAPS). VI. Distribution of the Small Organics HCN, C_2H, and H_2CO" Guzmán, Bergner, Law et al. ApJS, 257, 6 (2021)

49 "Molecules with ALMA at Planet-forming Scales (MAPS). III. Characteristics of Radial Chemical Substructures" Law, Loomis, Teague et al. ApJS, 257, 3 (2021)

50 "Long Baseline Observations of the HD 100546 Protoplanetary Disk with ALMA" Pérez, Casassus, Hales et al. ApJL, 889, L24 (2020)

51 "Dust traps in the protoplanetary disc MWC 758: two vortices produced by two giant planets?" Baruteau, Barraza, Pérez et al. MNRAS, 486, 304 (2019)

52 "Cm-wavelength observations of MWC 758: resolved dust trapping in a vortex" Casassus, Marino, Lyra et al. MNRAS, 483, 3278 (2019)

53 "An inner warp in the DoAr 44 T Tauri transition disc" Casassus, Avenhaus, Pérez et al. MNRAS, 477, 5104 (2018)

Submitted articles

54 "ALMA detection of continuum emission from GQ Lup B" Pirola, Facchini, Cugno et al., submitted to A&A.

55 "DEC/O: Benchmarking Protoplanetary Disk Chemistry Codes for C/H and C/O Retrievals" Anderson, Walsh, Aikawa et al., submitted to A&A.

56 "DEC/O I: A Chemical Census of 80 Planet-Forming Disks" Cleeves, Anderson, Guzmán et al., submitted to A&A.

57 "DEC/O IV. CO Visibility Modeling of Protoplanetary Disks: Gas Disk Sizes, Stellar Ages, and Radial Drift Signatures" Zallio, Rosotti, Vioque et al., submitted to A&A.

58 "The Disk-Exoplanet C/Onnection (DEC/O). VI. An Overview of Line Emission Radial Structure" Law, Cleeves, Green et al., submitted to A&A.

59 "DEC/O X: JWST and ALMA Reveal a Connection Between Inner- and Outer-Disk Carbon Chemistry" Colmenares, Bergin, Cleeves et al., submitted to A&A .

60 "DEC/O XI: Characterizing Formaldehyde (H2CO) emission and excitation properties in protoplanetary disks" Hernández-Vera, Guzmán, Miotello et al., submitted to A&A.

61 "Slowly Spiraling CQ Tau: Constraints on Disk Pattern Drivers and Accreting Companions" Tobin, Cugno, Meyer et al., submitted to ApJ.